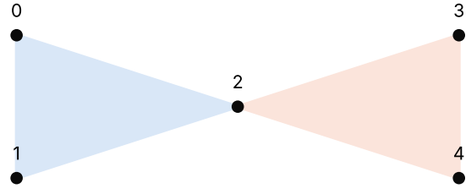
(a) the bowtie: windows $\{0,1,2\}$, $\{2,3,4\}$



$$\operatorname{rank}\begin{pmatrix} m_{012} & c_{01} & c_{02}c_{12} \\ m_{234} & c_{34} & c_{23}c_{24} \end{pmatrix} \le 1$$

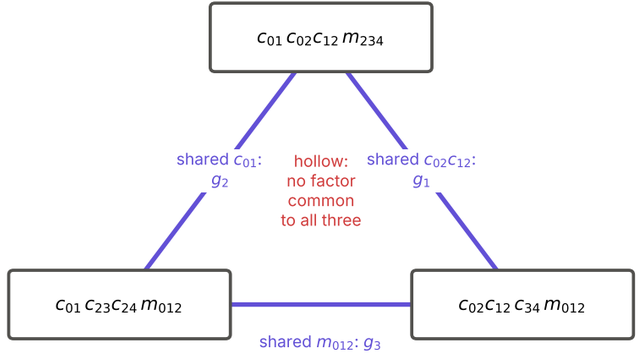
each column $\propto v_0v_1 : v_3v_4$; three minors = the generators;
the two rows = the two syzygies (Hilbert–Burch)

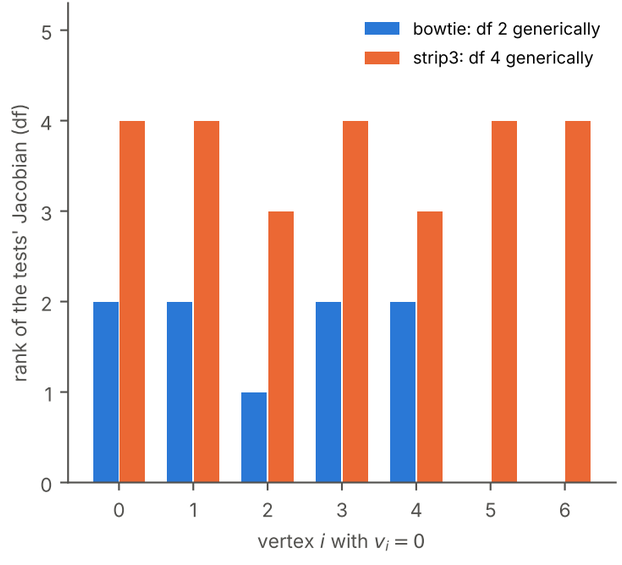
(b) a relation = a disconnected fibre

$b = v_0v_1v_2v_3v_4\,r$

no common factor: Δ_b = two disjoint edges,
$\tilde{H}_0(\Delta_b) = \mathbf{Q}$: a minimal generator
$c_{34}m_{012} - c_{01}m_{234}$ (the second-triple test)

(c) a syzygy = a loop of relations



$b = v_0^2v_1^2v_2^3v_3v_4\,r$: $\tilde{H}_1(\Delta_b) = \mathbf{Q}$, a first syzygy

(d) df drops only at a coincidence on a cut vertex
(bowtie: 2; strip: 2 and 4)